OBJECT ORIENTED SOFTWARE ENGINEERING

Ms. Archana A

Department of Computer Applications

OBJECT ORIENTED SOFTWARE ENGINEERING

Kanban

Archana A
Department of Computer Applications

Kanban

- The Kanban method is a **lean methodology** that describes methods for improving **any process or workflow**.

- Kanban is focused on **change management and service delivery**.

    - **Change management** defines the process through which a **requested change is integrated** into a software-based system.

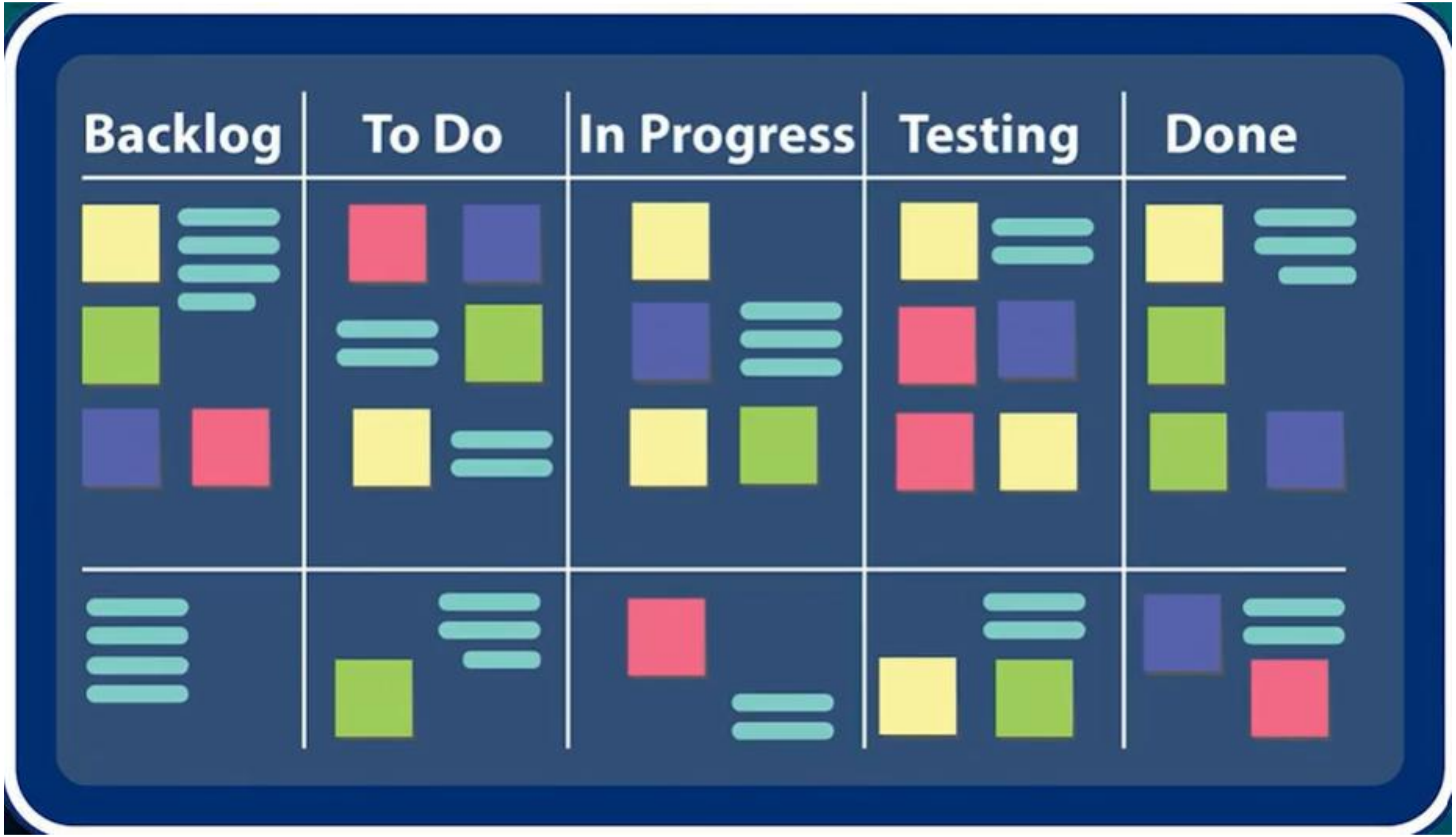
    - **Service delivery** is encouraged by focusing on understanding customer needs and expectations. The team members manage the work and are given the freedom to **organize themselves** to complete it.

- Policies evolve as needed to improve outcomes.

Kanban

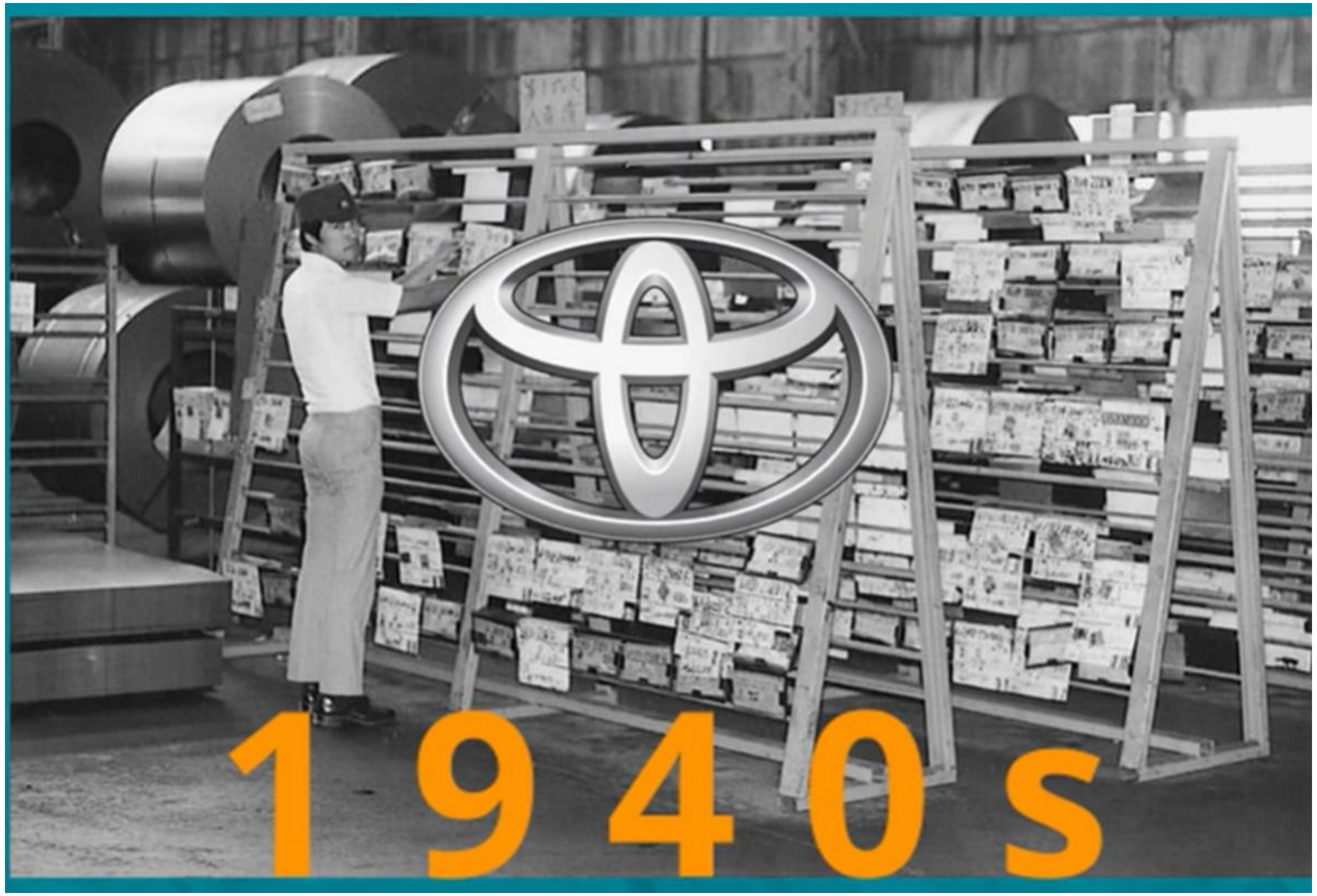
- The Kanban approach is a popular agile software development methodology that focuses on **continuous improvement**, **flexibility,** **and** **enhanced workflow.**
- It emphasizes visualizing the entire project workflow on a **Kanban board**, which **helps teams track progress, identify bottlenecks, and make data-driven decisions** to optimize their processes.

Fig: Kanban board



Kanban

- Kanban originated at **Toyota** in 1940, as a set of industrial engineering practices and was adapted to software development by David Anderson in 1950.



- Kanban **core practices** are as follows:

1. **Visualize Work:** Represent work items as **cards or tickets** on a **Kanban board**, moving them through columns that represent different stages of the workflow.

2. **Limit Work-in-Progress (WIP):** Set limits on the number of tasks that can be in each stage to prevent overburdening and maintain focus.

- **Manage Flow:** Optimize the movement of work through the workflow by identifying and removing bottlenecks.

- **Make Policies Explicit:** Clearly define and document policies that govern the workflow, such as WIP limits and prioritization rules.

- **Continuous Improvement:** Regularly **review and refine** the Kanban system based on **feedback and data analysis**.

Key Elements of Kanban

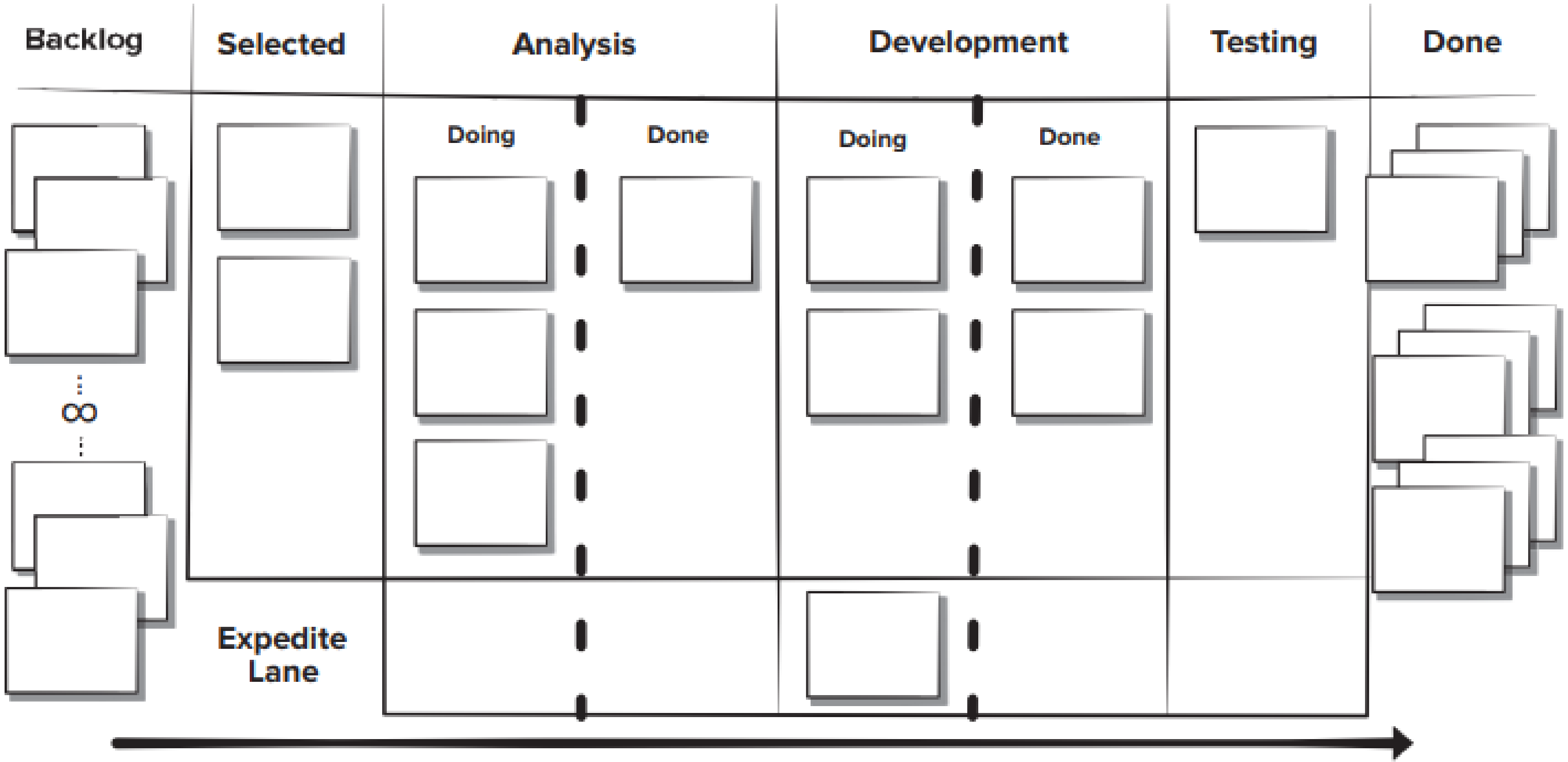


Fig: Kanban board

Key Elements of Kanban Implementation:

Kanban Board: A physical or digital board that visually represents the workflow, typically divided into columns like **"To Do," "Doing," "In Review,"** and **"Done."**

Kanban Cards: Cards represent individual tasks or work items, containing information like task description, priority, and due date.

WIP Limits: Maximum number of tasks allowed in each stage of the workflow, preventing overburdening and ensuring focus.

Swimlanes: Optional horizontal divisions on the Kanban board to represent different categories of work, such as by feature or team.

.

Kanban

- An "expedite lane" on a Kanban board is a **special lane reserved for the most urgent, high-priority work items** that **need immediate attention** and **can bypass normal workflow stages.**
- It represents a specific "Class of Service" for work with a high cost of delay, such as a **critical system failure** or a **request from a major client.**
- Tasks in this lane are handled first and may even exceed a team's typical Work-in-Progress (WIP) limits. .

- The **team meetings** for Kanban are like those in the **Scrum framework**.

- If Kanban is being introduced to an existing project, not all items will start in the backlog column.

- Developers need to **place their cards in the team process column** by asking themselves:
    - **Where are they now?**
    - **Where did they come from?**
    - **Where are they going?**

Kanban

- The basis of the **daily Kanban standup meeting** is a task called **"walking the board."**

- **Leadership of this meeting rotates daily**. The team members **identify any items missing from the board** that are being worked on **and add them to the board.**

- The team tries to advance any items they can **do** to "**done**." The goal is to advance the **high business value items first**. The team looks at the flow and tries to identify any impediments to completion by looking at workload and risks

- During the **weekly retrospective meeting**, **process measurements are examined.**

- The team considers where **process improvements may be needed** and **proposes changes to be implemented**.

- Kanban can easily be combined with other agile development practices to add a little more process discipline.

Kanban – real time examples

Example 1: Basic Kanban Board for Task Management

Columns: Backlog → Up Next → In Progress → Done
Description: Tasks start in the Backlog, move to Up Next when prioritized, then to In Progress when work begins, and finally to Done when completed. Each task is represented as a card that moves across these columns as it progresses

Example 2: Customer Support Kanban Board

Columns: New Requests → In Review → In Progress → Resolved
Description: This board helps a support team track incoming service tickets. New requests are triaged in the first column, reviewed for prioritization, worked on in progress, and finally marked resolved once completed

THANK YOU

Ms. Archana A
Department of Computer Applications

archana@pes.edu

+91 80 6666 3333 Extn 392

PES UNIVERSITY